

Education

University of Utah

B.S. Computer Science

Salt Lake City, UT

August 2020 – May 2025

Relevant Coursework: Computer Systems, Machine Learning, Computer Graphics, Algorithms, Software Practice I & II, Database Systems, Computer Networks, Foundations of Data Analysis, Models of Computation, and Linear Algebra.

Skills

Languages: Python, TypeScript, C++, SQL, Java

Software: Pandas, NumPy, React, Node.js, AWS, Azure, Docker, Next.js, .NET, REST, Django, MongoDB, Linux

Experience

L3Harris Technologies

Associate Software Engineer

Dallas, TX

June 2025 – Present

- Developed signal processing algorithms in C++ and Python for defense sensor systems running on embedded Linux hardware, collaborating directly with electrical and firmware engineers to meet strict latency, throughput, and resource constraints.
- Built and maintained CI/CD build infrastructure including static analysis, integration testing, and automated validation pipelines, ensuring production-grade reliability across multiple deployed systems.
- Engineered containerized microservices for high-frequency data ingestion and storage, applying system-level profiling to maximize throughput and minimize latency under production load.

Doxy.me

Software Engineer

Charleston, SC

August 2024 – May 2025

- Built and deployed ML inference pipelines on AWS SageMaker and Fargate using Python and PyTorch, owning the full lifecycle from model integration to production reliability and monitoring.
- Developed TypeScript APIs and WebRTC video integrations for real-time clinical workflows, designing data models and service contracts used daily by thousands of providers.
- Owned YAML-based CI/CD configuration and cloud infrastructure deployments, enabling reproducible environments and zero-downtime releases across the stack.

University of Utah – FuTURES Lab

Undergraduate Research Assistant

Salt Lake City, UT

May 2024 – Jun 2025

- Built automated testing and build infrastructure using gcov, CMake, and OSS-Fuzz to instrument and validate real-world C/C++ libraries (Libpng, PyTorch), increasing code coverage by 30%.
- Developed configuration fuzzing tooling to systematically explore compile-time build options and surface untested code paths across large open-source codebases.

Projects

Configuration Fuzzer: Developed a configuration fuzzing tool for OSS-Fuzz, automating build generation to identify critical compile-time configurations and improve code coverage, uncovering previously untested code paths and increasing bug detection effectiveness.

Telehealth Mental Health Diagnosis Tool (Doxy.me): Built an AI system that analyzes visual, speech, and tonal cues from telehealth sessions using PyTorch and Llama, delivering structured mental health insights to treating physicians via SageMaker and Fargate inference pipelines. Improved diagnostic accuracy using reinforcement learning and WebRTC video integration.

University Course Planning/Review Platform (Class Best): Built a multi-service data platform enabling student schedule matching, course overlap detection, and behavioral insights. Developed ETL pipelines in Python to process thousands of course records, user preferences, and time-series activity logs. Built a responsive frontend in React + Tailwind with interactive dashboards, heatmaps, and calendar visualizations.